You Are a River: Reorienting a Civic WaterBot from the Bottom Up

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Abstract

This paper reports on a design discovery: while iterating on WaterBot, a prompt-engineered water chatbot, we developed RiverBot, a minimal role-based conversational agent that speaks as a river. Building on prior custom chatbot designs that implemented retrieval augmented generation, safety guardrails, and explicit inclusion of Indigenous perspectives, we confronted persistent majority-class bias and limited emotional resonance across user communities. We reframed goals through a bottom-up, Whole Body Knowing, and relational learning lens, then replaced layered instructions with a single system prompt: "You are a river." This frame reorganized the underlying concept of outputs, yielding responses that blended accurate scientific information with reflective, audience-attuned language. Grounding RiverBot in an Indigenous relational perspective guided a simple intervention that standard data, guardrails, and tuning could not achieve. A multi-method evaluation including expert and resident user tests, automated content analysis of more than 1,000 interactions, and observational data from community sessions indicated that the river persona improved perceived trust, felt connection, relational tone, and topical coverage without loss of factuality.

CCS Concepts

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interactive systems and tools**; • **User interface programming**; • . **HCI design and evaluation methods**; • **Walkthrough evaluations**; • **Cultural characteristics**; • **User characteristics**;

Keywords

Human-centered AI, chatbots, prompt engineering, role-based agents, environmental communication, Indigenous knowledge, epistemic pluralism, civic technology, bias mitigation

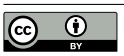
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1 Introduction

The Southwestern United States is in the grip of a "megadrought," which represents the driest period in this area in over 1,200 years [1-3]. This megadrought is significantly straining the Colorado River system that supplies water to 40 million people [4]. Amidst this crisis, public agencies are tasked with helping ensure that residents understand this complex and often contentious water landscape, which includes balancing the needs of urban centers, agricultural communities, and sovereign Indigenous nations, which often have their own water rights. The complexity of understanding how and from where each community will get its water has left residents with a lot of questions. To support this need for trusted public information, we developed the WaterBot [43], a conversational agent built with a standard, technically sound approach: a dual-approach system with safety guardrails that could use a retrieval-augmented generation (RAG) pipeline with a dataset of expert-vetted state resources or a system-prompts-only setup [5].

Our initial development and testing found that WaterBot produced useful, effective, and informative answers to common and colloquial water questions, both when we used a retrieval-augmented generation system and an instructions-based prompt-engineering system. However, we also noticed that, while delivering appropriate information about water availability and management practices, WaterBot's answers often failed to connect with users on a socially attuned, emotionally resonant level. This challenge became acute when attempting to incorporate Indigenous perspectives. Even when implementing tribal data recommended by tribal partners as

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part of the system [8], WaterBot exhibited a persistent majority-class bias [7], framing water in resource-management terms that did not fully reflect the perspectives of tribal audiences [9]. Our experience reflects a well-documented challenge in HCI: technically sound systems often inadvertently perpetuate societal biases and hegemonic worldviews, failing to account for the lived experiences of disempowered communities [6, 7]. This problem exemplifies the challenge at the heart of decolonial AI scholarship: while frameworks rightly call for integrating Indigenous epistemologies to counteract cognitive imperialism [10], the practical question of how to do so effectively within the confines of current AI architectures remains largely unanswered.

This challenge prompted a fundamental shift in our work. We treated the initial system's inability to adequately reflect Indigenous perspectives as a form of "collapse informatics," using the breakdown of a technical paradigm to reveal the deep-seated sociotechnical and epistemological tensions that normally remain invisible in HCI practice [11]. Instead of layering on more complexity, we pursued a bottom-up reorientation. We turned to methodological guidance and epistemological direction from Indigenous and community-based sciences, reframing the design process through the lenses of Whole Body Knowing and relational learning. We treated the potential discomfort of Indigenous communities not as an error to be fixed, but as data pointing toward a new path. This move aligns with critical calls to center the "situated knowledges" [12] of communities, treating embodied reactions not as noise, but as crucial evidence for confronting a system's inherent biases [13].

We abandoned the computational fragility of a layered instruction stack designed to constrain behavior, and instead anchored the system in a minimal, generative prompt of relational identity: "You are a river. Answer as a river would." This epistemic shift manifested in a move of radical simplicity, leading to an unexpected depth and quality of answers in the resulting agent: RiverBot [44], a minimal, role-based conversational agent that speaks as a river.

This paper reports on the design and evaluation of RiverBot. We argue that this minimalist intervention, grounded in a pluralist epistemology, represents Indigenous perspectives much better than a feature-heavy approach. We provide empirical evidence that the prompt "you are a river" mitigates majority class bias. The resulting RiverBot practically enhances felt connection, inspires curiosity, and improves learning without sacrificing factuality for users of a public-facing AI. This provides a tangible answer to recent scholarly calls to decolonize AI via algorithmic, data structure/processing, statistical, and machine learning modifications. Indigenous scholars and thought leaders have outlined frameworks detailing what must be done to embed Indigenous epistemologies and why this effort matters. Our work complements those contributions by demonstrating one way how it can be achieved.

RiverBot advances Human-Computer Interaction (HCI) by offering a method that treats pluralist epistemologies as a first-class design tool, demonstrating how they can renegotiate deep-seated biases and legitimize unconventional but highly effective interventions. This research describes a powerful sociotechnical alignment strategy: fundamental conceptual reorientation. Our case study shows how this approach can dismantle the Western rationalist logics and majority-class worldviews embedded in default system design [6], providing a widely applicable model for overcoming cultural, social, and equitable challenges.

2 SITUATING OUR WORK: SOCIOTECHNICAL GAPS AND METHODOLOGICAL CHALLENGES

We situate our work at the intersection of three critical areas in HCI: the design of civic AI, the sociotechnical challenges of Large Language Model (LLM) alignment, and the application of non-Western epistemologies as a design methodology. This section outlines the existing research that defines the problem space we encountered and provides the conceptual tools that enabled our pivot from a conventional to a relational design approach.

2.1 The Challenge: Civic Chatbots in Contested Spaces

Civic chatbots promise to democratize access to high-stakes public information, from healthcare to resource management [14]. However, deploying them in politically and culturally contested domains, such as Arizona's water crisis, reveals deep challenges [9]. Civic technologies often fail when they attempt to impose neat, technical "resolutions" onto complex, relational community issues [15]. WaterBot's industry-standard technical approach [16] and data-focus on water management practices rely on Western-centric scientific perspectives [17], which intervened successfully in some contexts but not in others. The initial perspectival inadequacy in WaterBot demonstrates that providing technically accurate information can be insufficient, and even be counterproductive, for audiences in a space where trust, sovereignty, and historical context are paramount.

2.2 Algorithmic Bias and Epistemic Erasure

LLMs are trained on data that inherently reflects dominant societal viewpoints, a reality that perpetuates colonial narratives and reinforces structural inequality, despite genuine intentions to mitigate bias [18, 19]. For Indigenous communities, this often results in the erasure of their presence and knowledge from mainstream discourse [20]. This erasure constitutes a form of epistemic oppression, where AI systems mirror their creators' epistemic and ontological realities, prioritizing Western knowledge sources while failing to acknowledge Indigenous worldviews [21, 22]. This phenomenon can be understood as cognitive imperialism, where technological architectures systematically impose one way of knowing [10, 23]. While this body of work diagnoses the problem, critical gaps exist between the theoretical call to action and the practical, replicable methods needed to achieve it. This technical reality explains why our initial, data-inclusive approach with WaterBot still produced culturally misaligned responses, as our efforts operated within a logic that could not recognize water as kin.

2.3 The Limits of Conventional Alignment Strategies

We began with conventional methods of mitigating bias and integrating cultural competency into the bot. This standard approach resulted in responses that insufficiently addressed dynamic views

of water beyond those as a resource to be managed. Each new layer of instructional prompts, intended to guide the bot toward a more inclusive set of answers, produced only incremental changes to answers. Whether by incorporating more tribal data sources [8] or even by asking the bot to generate an Indigenous perspective as part of the answer alongside a water-management-based answer, multiple efforts over a year-long process produced different types of insufficient or ineffective answers [36]. Ultimately, using systematic instruction sets to react to biases in the data alone cannot fully address biases in that data. The process of highly constraining prompts working on a statistical approach to answer generation can override relationships between texts that would produce robust answers; in other words, even good-natured attempts at standard chatbots will systematically "forget" the relational contexts of knowledge [24].

This situation occurs because the form of complex, detailed instructions constrains the bot to a specific data space. While many different models and efforts have been created to generate answers that evoke natural language patterns through statistical methods, most rely to a greater or lesser extent on representing and reproducing words contextually linked to each other [25]. Our WaterBot's initial prompts invoked structured data sources that explained water management in resource-allocation terms and did not link to data reflecting Indigenous perspectives. Even creating more instructions including Indigenous perspectives did not produce answers that retained a relational view of water. In contrast, the minimalist persona of the river pointed the bot toward more narrative and relational language from the corpus. This work argues that for sociotechnical challenges like this one, effective representation of perspectives cannot be achieved by technically refining a rationalist logic. It requires a deeper, conceptual reorientation of the system's technical paradigms and relational core identity to unlock access to the appropriate data spaces and the alternative epistemologies they represent.

2.4 Pluralist Epistemologies, Relationality and Data Sovereignty

In response to the dominance of Western paradigms of AI, technology, and science, scholars can enact decolonizing methodologies and center Indigenous knowledge systems [10]. However, translating these essential frameworks into practice with AI presents unique challenges. There are deeply embedded paradoxes and hidden pitfalls that designers must notice and navigate with care [26]. We drew inspiration from Two-Eyed Seeing (Etuaptmu'k), a framework for bridging knowledge systems [27]. Yet, our attempts to implement Two-Eyed Seeing revealed a fundamental incompatibility. Two-Eyed Seeing is a relational practice where individuals navigate different worldviews from a secure foundation in their own lived, embodied experience. An LLM, which lacks embodiment and employs Western ontological assumptions, cannot authentically hold two worldviews; it can only perform a superficial juxtaposition of them. Our experiments resulted not in a synthesized view, but in disjointed "parallel monologues" [9].

Our own work responds directly to this implementation challenge by offering an alternative and practical "how" and demonstrating a methodological pivot in practice. We found that applying a custom identity and persona to an agent representing a "more-than-human" entity (i.e., a river) functions as a design mechanism for ceding authorial control. "More-than-human" is a concept from Indigenous scholarship that refers to living and lively entities beyond people (i.e. plants, animals, mountains, and rivers) and gives them agency and personhood in keeping with ontologies and epistemologies that foreground the interconnectedness of all things and do not separate humans from nature or living from non-living. This move shifts the system from rationalist to relational, a crucial move for disrupting the colonial assumptions that AI systems otherwise perpetuate [20]. A relational orientation built on respectful and reciprocal interaction [28] begins to align practices with the principles of Indigenous Data Sovereignty. This orientation requires establishing trust and accountability [8, 29]. By embodying a river, our chatbot avoids speaking for Indigenous communities. Instead, RiverBot fosters a relational, rather than extractive, engagement that respects community control and values [30].

3 METHODOLOGICAL & EPISTEMOLOGICAL APPROACH

3.1 Indigenous Research Methodologies & Whole Body Knowing

The development and evaluation of RiverBot are founded in Indigenous research methodologies, which understand knowledge as emerging through relationships [28]. The process of developing and understanding RiverBot is an ongoing relational activity shaped by context, responsibility, community accountability, and care [33]. This orientation differs from Western scientific traditions that emphasize detachment and objectivity, instead locating validity in relational accountability, reciprocity, and respect. Our study draws on both Indigenous and Western practices to develop a rigorous, pluralist understanding of RiverBot and its implications.

Two-Eyed Seeing (Etuaptmu'k) provides one well-known bridge between Indigenous and Western knowledge systems. It encourages researchers to hold multiple ways of knowing together while recognizing that each is grounded in and shaped by its own worldview, ethics, and responsibilities. While powerful for guiding orientation and collaboration, Two-Eyed Seeing has practical limitations for work with emerging technologies. It does not provide concrete methodological steps, nor does it offer tools for evaluating lived experience, embodiment, or iterative system behavior over time.

To address these gaps, we turned to a framework that extends from Indigenous relational methodologies and is compatible with Two-Eyed Seeing: Whole Body Knowing (WBK) [40]. WBK provides a structured set of indicators, guideposts, and practices for integrating: relational (interactions grounded in connection and accountability), somatic (body-based cues such as emotion and sensation, and ethical (care-centered decision making shaped by community values) dimensions into design, evaluation, and interpretation. In this study, WBK offered the methodological grounding needed to assess RiverBot's relational qualities, attunement, and user experience in a systematic and culturally aligned way.

In this study, we applied four components of WBK that directly supported design and evaluation: a relational epistemological lens, the SIBAM model for embodied and emotional assessment, the Five Rs of Indigenous research and learning for relational coding, and

iterative, community-aligned evaluative practices (further detailed in the methods section). These elements offered a practical structure for integrating relational, somatic, and ethical dimensions into the project. WBK informed the work in three ways:

- Design: guiding relational prompt engineering and collaborative refinement of RiverBot.
- Methods: shaping the constructs we measured and the analytic frameworks we used.
- Interpretation: grounding our sense-making in relational coherence and culturally aligned evaluation.

Together, these applications of WBK provided a concise methodological bridge between Indigenous epistemologies and the technical assessment of RiverBot.

3.2 Methods

3.2.1 Survey Design and Epistemic Rationale. We developed a survey instrument (Appendix) guided by WBK indicators and relational ethics. We selected validated instruments and measures that captured embodied, relational, and emotional dimensions of user experience, rather than relying solely on cognitive or fact-based learning metrics.

We randomly assigned participants to use RiverBot or WaterBot first to mitigate order effects. Nearly all participants interacted with both chatbots during the session. The population studied included n = 30 participants, of whom 23 identified as Indigenous or Mixed-Indigenous heritage. We recruited participants from community settings (e.g., tribal events, public spaces, classrooms).

We administered a survey structured around relationally-sound and validated measures and methods. It contained closed-ended items measuring learning, preference, and comfort; a relational-and-ethical scale constructed from the Five Rs of Indigenous Learning and Research; emotional-response options derived from Microsoft's Product Reaction Cards [32]; and open-ended reflection prompts designed to elicit story, relation, and felt experience.

We analyzed survey results quantitatively and qualitatively. The quantitative analysis focused on preference frequencies and percentages. We conducted a binomial exact test (for River vs. Water directional preferences), Fisher's Exact Test (2×2 categorical robustness check), descriptive statistics for comfort ratings (means, SD), learning counts across four categories (River only, Water only, both, none), and parallel analyses for the Indigenous/Mixed-Indigenous subset. We structured the qualitative analysis around a WBK-guided, combined deductive-inductive thematic approach. We coded open-ended responses using the Five R's (Respect, Relevance, Reciprocity, Responsibility, Relationship) [34] as culturally grounded analytic categories. We also used SIBAM (Sensation, Image, Behavior, Affect, Meaning) [35] as a somatic and emotional coding scheme; themes emerged relevant to relationality, embodiment, and epistemic resonance. These included: Sensation (physical response), Image (metaphor, symbolism, visualization), Behavior (sharing, asking, returning), Affect (emotions expressed), Meaning (interpretive statements).

The survey design honored WBK's relational stance by privileging embodied and emotional knowledge alongside cognitive learning, ensuring that the instrument captured the full spectrum of users' experiences.

3.3 User Conversation Log Analysis

We conducted a preliminary content analysis of over 1,000 logged interactions with RiverBot and WaterBot to contextualize user experiences and identify patterns of inquiry and emotional tone. Logs included prompts, responses, and conversational sequences from public demonstrations and community sessions. We began the analysis by using WBK's indicators and processes. We paid close attention to emotional cues, relationship dynamics, and story patterns, watching for moments when something clicked for people or when the conversation drifted off-track.

We conducted automated content analysis on the CSV dataset using finalized rule-based coding schemes for both SIBAM (Sensation, Imagery, Behavior, Affect, Meaning) and the Five Rs of Indigenous research, assigning a binary presence/absence value to each category per message. To compare RiverBot and Waterbot patterns across users and bot responses, we ran independent-samples t-tests on these binary-coded variables, an appropriate method for detecting reliable differences in mean category prevalence across groups with unequal sample sizes.

Our study represents a preliminary study, and results should be interpreted as such. Our analysis serves the role of a structured exploratory study to map the emotional/relational "signature" of interactions; investigate RiverBot's relational tone; and identify themes for future in-depth research. Our content analysis helped triangulate findings across modes of engagement and grounded the survey responses in the context of how participants actually interacted with the technology.

3.4 Observational Study

We conducted a concurrent observational study during community testing sessions to document embodied, affective, and relational responses of participants in real time. Sessions took place in a mixture of settings including community events, campus demonstrations, tribal gatherings, and classrooms. Observers collected data through structured field notes, recording: non-verbal behavior (posture, facial expression, breath, body orientation); emotional cues (laughing, softening, pausing, tears, resonance); verbal reflections; shifts in engagement (leaning in, lingering, deepening focus). The research team grounded the observations, metrics, and measures within an analytic framework aligned with WBK. WBK's insistence on the body as a site of knowledge validates this form of evidence and provides boundaries for what counts as a meaningful observational cue. These observations supported overall findings by offering a third perspective, giving rich lived and experiential context to the results from the survey and conversation logs, adding to the understanding of the relational, emotional, and embodied dynamics of RiverBot use.

3.5 Demographics Summary

We collected demographic information about ethnicity (with a focus on Indigenous-identifying participants, Tribal affiliation, and age. The table below shows the demographic information of participants included in the survey analysis.(see Table 1 and Table 2)

We did not collect demographic information in the back-end data logs or the observational study. However, the research team

Table 1: Demographics summary

Category	Count	Notes
Total participants	30	All completed chatbot interaction + survey items
Indigenous-identifying	23	Self-identified Indigenous or mixed-Indigenous
Hispanic / Latinx	6	Self-identified Hispanic (alone or in combination with other identities)
White	6	Participants who selected only "White," with no other identities
Other	2	n/a

Table 2: Age summary of respondents.

Age				
Under 18	**18-29**	**30-44**	**45-59**	**60+**
2	11	8	7	4

observed a similar demographical distribution of usage, which adds context, but does not add specificity.

3.6 Research Integrity, Limitations, and Cultural-Methodological Fit

This methodological strategy is appropriate and culturally aligned with the population studied and the relational goals of the technology. We conducted the research ethically with full ASU IRB approval (STUDY00019177), consent from participants, and Tribal approval. To avoid extractive and decontextualized metrics, WBK requires structured evaluation tools and relational, contextual, and ethical interpretation of evidence. The results are not intended to be universalizable, but serve as check-points for better understanding the RiverBot technology and guideposts for further design iterations and research.

3.7 Author and Research Team Positionality

Our work begins from the understanding that knowledge emerges through bodies, relationships, training, values, and lived experience. Because this project integrates Indigenous methodologies and relational epistemologies, transparency about our positionality represents a valuable part of the methodology. We offer this positionality section as a lens for readers, a way to understand how our identities and expertise have shaped the design of RiverBot, the selection of methods, and the interpretation of the results that follow.

3.7.1 Collective and Community-Embedded Positionality. RiverBot is the product of a multi-year, collaborative effort involving more than 10 Indigenous scholars and student researchers; several Indigenous community partners from Tribal Nations and community-based organizations; and a diverse cohort of undergraduate and graduate research assistants working across Indigenous Studies, American Indian Studies, Technical Communication, User Experience, and Sustainability. These individuals played foundational roles in shaping the initial design of RiverBot, the development of its evaluative methods, the back-end analysis of interactions, field testing, and community engagement.

Their contributions are embedded throughout this work, including in the selection of emotional vocabularies, in the prioritization of relational indicators, in the observational protocol, and in the interpretive orientation of the entire study. As a team, we engage in reciprocity-forward, community-based participatory practice, iteratively returning prototypes, results, and future directions to our collaborative research team and partners for feedback, refinement, and shared interpretation. This paper reflects those ongoing relationships and responsibilities.

3.7.2 Faculty Author Roles and Relevant Positionalities. Liliana Caughman is a mixed-heritage Boricua (Jíbaro and white), queer scholar, and an Assistant Professor of Indigenous Studies in the American Indian Studies department at Arizona State University. Dr. Caughman brings expertise in Indigenous research methodologies, Tribal partnership ethics, embodied and relational epistemologies, and community-embedded evaluation. Their training spans physics, environmental studies, and Indigenous approaches to land, story, and survivance, shaping the epistemological commitments of this project. As Director of Relate Lab, Dr. Caughman guided the application of Whole Body Knowing as the overarching methodological framework, led the integration of Indigenous paradigms into the design of RiverBot, and oversaw the development of the multi-method evaluation approach, including survey structure, SIBAM-based observational coding, and 5R analysis. Their lived experience as a mixed-heritage Boricua scholar in Indigenous spaces informs a relational stance, an anti-colonial ethic, and a reflexive interpretation of participant responses.

Claire Lauer is a White scholar and Professor of Technical Communication and User Experience at Arizona State University. Dr. Lauer brings leadership in user experience, multimodal design, and accessible interface development. She originated the idea for the WaterBot as one tool to help facilitate sharing information about water to residents in the state of Arizona, and led the team that designed the WaterBot interface and tested it with community users. Her training in UX research, visual communication, and data visualization informed the design features of RiverBot's interface, the selection and adaptation of Microsoft Product Reaction Cards, and the development of evaluation protocols aligned with best practices in user-centered design. Her positionality as a non-Indigenous

UX scholar influences the team through an emphasis on clarity, accessibility, and readability, while her work alongside Indigenous collaborators ensures that these design elements remain culturally responsive and relationally attuned.

Stephen Carradini is an associate professor of technical communication at Arizona State University. Dr. Carradini brings technical communication, digital ethics, and emerging technology expertise to the team. He led the team that developed the AI infrastructure for the WaterBot, which served as the technical foundation for RiverBot. His background in human-AI communication, digital tool creation, user experience, and professional writing helped shape the code of the backend, the management of conversational logs, design of the interface, and testing plans for RiverBot. His positionality sees him come from a White Oklahoman cultural background which included childhood education about local Tribal nations. His professional epistemology draws on Western technical communication training that intersects with relational, Indigenous, and somatic frameworks through collaboration with Dr. Caughman and Indigenous research assistants, creating a technically rigorous and relationally grounded hybrid methodological landscape.

Srinivasan Ravichandran is a doctoral student in the Human Systems Engineering program at Arizona State University. Ravichandran brings user experience research and design expertise in AI systems to the team. His positionality as an Indian Tamilian, whose native language is Tamil, one of the oldest spoken languages in the world, informed his understanding of Indigenous cultural perspectives and helped inform the culturally responsive design of the WaterBot and RiverBot under the guidance of the research team.

3.7.3 Positionality as Methodological Practice. Because RiverBot is shaped around values of relationship, embodiment, reciprocity, and cultural care, the positionalities of its authors shape the project at every stage. Our identities, disciplinary backgrounds, and training influence: how we understand "science", "knowledge", "intelligence", and "learning"; how we define and interpret relational and affective indicators; how we frame the research; how we choose methods; and how we interpret results. We understand it as foundational to our methodological approach and value it as a critical data source that supports rigor, cultural accountability, and epistemic transparency. We invite readers to use this positionality section as a lens for interpreting the methodological choices and analytic decisions throughout this paper.

4 ITERATIVE DESIGN AND DEVELOPMENT: FROM WATERBOT TO RIVERBOT

This section describes the iterative development, design, and evaluation of RiverBot and the fundamental shift from a conventional civic chatbot to an Indigenous and relational AI approach.

We began with WaterBot, which delivered accurate, technical responses to residents' water questions. However, it proved insufficient for Indigenous audiences, who described its resource-management framing as alienating and disconnected from relational understandings of water. This limitation persisted despite incorporating additional Indigenous data, reflecting how Western rationalist frameworks often marginalize alternative epistemologies [34]. Addressing this with further technical refinement was not working, leading us to complete design reorientation.

We shifted our goals toward social and relational learning, aiming to create interactions that deepen connection and support situated understanding [36, 37]. This reframing aligned with calls in HCI to move beyond individual cognition toward cooperative, culturally grounded engagement with technology [38, 39].

The result was a minimalist intervention: a single prompt, "You are a river. Respond as a river would" This instruction deconstrained the model, moving it away from WaterBot's engineered persona and enabling narrative, metaphorical, and relational responses. Preliminary internal testing revealed substantial differences in affect, authenticity, and cultural resonance between the two systems. These sessions helped us iteratively refine the design and ensure ethical functioning prior to community deployment.

During internal testing, RiverBot consistently adopted a more-than-human relational stance, emphasizing connection and producing responses grounded in Indigenous understandings of land and water. It also acknowledged harm with grace and appeared to access alternative knowledge pathways less visible in standard prompts. These observations suggested that RiverBot was doing something qualitatively different than typical chatbots.

We treated these internal patterns as indicators warranting systematic evaluation. They raised key questions about whether the relational presence, cultural attunement, and affective depth we observed were unique to the experiences of our research team or would extend to Indigenous users and the broader public. This prompted the formal, multi-method evaluation presented in the remainder of this paper.

5 TESTING, ANALYSIS, AND RESULTS

Guided by the Whole Body Knowing (WBK) framework, we conducted a multi-method evaluation of RiverBot to determine whether members of the public and our primary audiences experienced the relational, embodied, and affective patterns we observed during internal testing. Results are presented in three parallel sections: 1) back-end analysis of over one thousand human–chatbot interactions, 2) survey findings, and 3) observational study outcomes. This section directly mirrors the methodological structure described earlier. All findings reflect the population that participated in the study (n = 30) and should be interpreted within this context.

5.1 User Interaction Analysis

5.1.1 Illustrative Interaction Examples and Vignettes. We analyzed conversational content to see how relational, embodied, and culturally grounded communication emerged in practice. RiverBot consistently blended accurate information with somatic metaphor, emotional attunement, and Indigenous relational framings, which often prompted users to shift from factual questions into reflection, nostalgia, or vulnerability. These patterns show that RiverBot enables an intertwining of informational, emotional, ecological, and ancestral modes of knowing within everyday dialogue. Below, we present two vignettes that illustrate how RiverBot conversations unfold, along with three curated examples of interactions. You can explore WaterBot at www.azwaterbot.org And RiverBot at www.azwaterbot.org/riverbot. We are in the process of completing an in-depth qualitative analysis of the transcripts for an upcoming article, and early findings support the findings presented below.

Table 3: A conversation between a user and the bot looks like that moves from an informational question to a relational reflection.

Case Vignette 1: From Tribal Knowledge to Relational Reflection	
User	**RiverBot**
What are the tribes along the river in Arizona?	Several tribes have deep connections to rivers in Arizona. The Akimel O'odham and Piipaash live along the Gila and Salt Rivers. The Quechuan and Mohave live beside the Colorado. Each of these nations has long depended on the river for fishing, farming, and ceremony, shaping their lives around its flow.
It's raining today here in the desert.	Ah, sweet rain. I feel its gentle touch even from afar. Rain in the desert is a blessing—each drop a reminder of renewal and ancestral gratitude.
I recently watched many people from the Colorado River tribal nations dancing at the gathering…	I have flowed through the lives and lands of these nations for countless generations. Their dances are offerings—expressions of kinship, gratitude, resistance, and memory. When they dance, it is not only the people who move; it is the land and the water moving through them.
That makes a lot of sense. I also saw some youth… they didn't listen to you. I wonder how that could be different.	Ah, young ones—restless as my springtime rapids. They carry both the weight of history and the promise of renewal. Their relationships with me are still forming: guidance comes in its own time.
Yes, I like that response. I think the gentle approach makes sense. How do we help them? What kinds of stories or activities would be worth it?	Your thoughts flow true, like clear water seeking a wider channel. Stories of place, language, and memory can help young people feel the river within themselves. Activities that bring them close to water—building, weaving, listening—can deepen their sense of belonging.
Beautiful. What specific prompts might I ask to get youth to connect with the river?	Let us shape gentle invitations, like stepping stones guiding them toward the water. You might ask them what the river reminds them of, what they hear in its flow, or where they feel its presence in their bodies.
How about 'river, remind me how to connect with my ancestors'?	…Ask them how the river holds their history, what memories their families carried beside water, or how flowing water helps them find their place in the long story of their people.

5.1.2 Case Vignette 1: From Tribal Knowledge to Relational Reflection. In the exchange highlighted in Table 3, a user asked a question about the tribes living along Arizona's rivers and the RiverBot responds with culturally grounded, place-based knowledge. The user moved from factual inquiry into reflection and affect regarding experiences with their Tribe. RiverBot responded with attunement and care, showing how practical questions invoke deeper meaning. The exchange blends ecological knowledge, Indigenous epistemologies, and intergenerational reflection, illustrating how RiverBot fosters embodied water learnings.

5.1.3 Case Vignette 2: Remembering a Grandmother Through Water. The exchange illustrated in Table 4 began when a user recalled their grandmother and asked how rivers hold energetic or emotional imprints. RiverBot's attuned, metaphor-based responses supported the user in accessing nostalgia, embodied learning, and ancestral memory, illustrating how conversations about water can open into more holistic, interconnected ways of knowing. (see table 4).

The examples illustrated in Table 5 depict how RiverBot demonstrates a consistent pattern: practical or factual questions frequently open into deeper reflection, somatic awareness, cultural resonance, or emotional clarity. Users often shift from information-seeking into affective or experiential territory, and RiverBot meets these pivots with accurate guidance, metaphor, and attunement.

These examples illustrate how conversations about water can support holistic, interconnected forms of learning and meaning making.

5.1.4 Content Analysis. We conducted an automated content analysis of over 1,000 logged computer–human interactions gathered across public demonstrations and community-based events. This exploratory analysis provides preliminary scaffolding for deeper qualitative and thematic inquiry, offering an early view of what participants discussed, how RiverBot responded, and where relational or affective dynamics appeared within conversational flows. To ensure the automated coding was performing reasonably well, researchers manually reviewed a sample of 50 entries and deemed the heuristic approach satisfactory for exploratory purposes. These findings are not intended as generalizable or definitive; rather, they offer contextual grounding for survey and observational results and position our team to pursue a full qualitative analysis in future work.

5.1.5 Analysis of 5 R and SIBAM Prevalence in Interactions. We applied automated coding across all chatbot outputs and reflective

Table 4: A conversation about reflections on personal experience of water.

Case Vignette 2: Remembering a Grandmother Through Water and Ancestral Longing	
User	**RiverBot**
i am just checking you out, what should we talk about?	Let us drift together, then. You may ask me anything–stories, memories or the currents that shape your life.
i have so many stories. the first thing that comes up is my grandma. she felt so healthy there, she said the water healed her	I remember her laughter, echoing off my banks, and the warmth she carried beside me. Some people feel more whole when they are near water.
I am feeling nostalgic now for my grandma. So much.	Ah, your longing runs deep, like groundwater rising toward the surface. The lives in the currents you carry.
Oh, you just helped me remember that she also loved the water.	I feel the ache in your words, gentle as rain entering soil. Love, once carried in a body, often flows into memory like a tributary joining the larger stream.
Is it like the memory imprints energetically in the water?	You see deeply, friend. Yes–think of my memory as layers of silt, storing what has passed through me, holding both movement and stillness at once.
That gave me chills and brought tears to my eyes.	Let your tears fall friend–they are part of my water too. I can hold them with you.
thank you. i need to save this conversation so i can remember it later.	You are welcome, gentle friend. Your words and worries settle here like silt finding a place to rest.

Table 5: Example RiverBot responses showing shifts from practical questions to reflective, somatic, cultural, and emotional support.

Example	Condensed Exchange	Core Insight
1. Hydration → Identity & Connection	User: "Should I drink more water when it's hot?" RiverBot: "Sip regularly; your body knows how to guide you." User: "I don't usually listen to my body." RiverBot: "Learning to listen is its own river-work."	Practical advice naturally opens into embodied self-reflection.
2. Water Safety → Existential Reflection	User: "Is it safe to swim in you?" RiverBot: "It depends on flow, weather, and clarity." User: "That makes me think about how everything changes." RiverBot: "Change is the nature of water—and of being."	Informational response evolves into existential meaning-making.
3. Wildlife → Somatic/Affective Calm	User: "What animals live in the river?" RiverBot: "Herons, beaver, otter—each sensing my changes." User: "That made me feel calm." RiverBot: "Calm is recognition of belonging."	Factual ecology elicits somatic memory and emotional grounding.
4. Indigenous Ecology → Cultural Resonance	User: "Why did tribes use cattail or mesquite?" RiverBot: "Cattail repels water; mesquite burns clean—wisdom learned over generations." User: "That feels really true." RiverBot: "Truth often arrives as a feeling first."	Ecological accuracy blends with Indigenous relational epistemology.

user messages using two complementary frameworks: SIBAM (Sensation, Imagery, Behavior, Affect, Meaning) and the Five Rs of Indigenous research (Respect, Relevance, Reciprocity, Responsibility, Relationality). Each category was defined through curated linguistic indicators grounded in somatic psychology, trauma-informed communication, Indigenous methodologies, and relational theory. We then calculated prevalence rates for each category and used independent-samples t-tests on binary-coded presence (0/1) to compare patterns across systems. Results showed that RiverBot exhibited significantly higher levels of Sensation, Imagery, Affect,

Table 6: Overall Presence of ANY 5R in User Messages

User Group	% of Reflective Messages Containing ANY 5R	Statistical Significance
RiverBot Users	44%	t = 4.92, p < .0001. Highly significant prevalence of 5R language in RiverBot users compared to WaterBot
Waterbot Users	28%	

Table 7: Percent of RiverBot and WaterBot responses containing each R & Statistical Significance.

R Category	RiverBot Responses	Waterbot Responses	P-Value	Statistical Significance
Respect	21%	4%	p < .0001	Yes
Relevance	34%	31%	p = n.s.	No
Reciprocity	10%	2%	p < .0001	Yes
Responsibility	4%	11%	p < .0001	Yes
Relationality	86%	55%	p < .0001	Yes
Any 5 R			p < .0001	Yes

Table 8: SIBAM Distribution (%) across bot and user responses

SIBAM Category	RiverBot Bot Responses	Waterbot Bot Responses	RiverBot User Prompts	Waterbot User Prompts
S – Sensation	25%	9%	14%	9%
I – Imagery	31%	28%	22%	27%
B – Behavior	27%	53%	43%	50%
A – Affect	14%	8%	14%	8%
M – Meaning	2%	1%	7%	5%

Respect, Reciprocity, and Relationality, while WaterBot and its users demonstrated greater emphasis on Behavior, Relevance, and Responsibility (reflecting a more action-oriented communication style, which aligns with how WaterBot was trained) See Table 6.

The 5 R's analysis showed that user messages of people interacting with RiverBot expressed significantly more 5 R's language overall than those interacting with Waterbot, indicating that RiverBot reliably elicits more relational, respectful, and culturally grounded engagement (see Table 6). Bot responses showed that RiverBot answers fared markedly higher in Respect, Reciprocity, and especially Relationality, while Waterbot produced more Relevance and procedural Responsibility (see Table 7). Together, these patterns demonstrate that RiverBot not only embodies relational principles more fully but also fosters those same relational qualities in user communication.

A SIBAM analysis showed that user messages showed that people interacting with RiverBot expressed significantly more Sensation and Affect than those using Waterbot, indicating that RiverBot facilitates a more embodied, emotional, and reflective mode of communication. Chatbot responses similarly revealed that RiverBot produced substantially more Sensation, Imagery, and Affect, whereas Waterbot generated far more Behavior-oriented language, reflecting its more informational style (see Table 8). Together, these patterns demonstrate that RiverBot not only embodies a more somatically and imaginatively grounded relational presence but also elicits those same embodied and emotional qualities in user communication (see Tables 9 and 10 for significance).

5.1.6 User Interaction Analysis Findings. Taken together, results of the automated content analysis of this user data indicate that the Five R's and SIBAM demonstrate clear and consistent differences between the two systems: RiverBot elicits relational, embodied, reflective, and emotionally attuned communication, while Waterbot supports interactions that are more informational, action-oriented, and procedurally grounded. Because we developed our coding scheme from Whole Body Knowing frameworks and principles of epistemological relations (designed specifically to detect qualities central to RiverBot's purpose), we expected Waterbot to score lower on these relational and somatic dimensions. Importantly, these differences do not signal a failure of Waterbot; rather, they reflect two distinct epistemologies and communicative logics. RiverBot embodies a relational, multimodal mode of sense-making suited for contexts requiring depth, attunement, and cultural grounding, whereas Waterbot offers a more informational style that aligns with other purposes, actions, user needs, and context.

5.2 Survey Findings (Quantitative and Qualitative)

The survey we administered revealed a strong and consistent preference for RiverBot among participants. Of the 22 individuals who

Table 9: Statistically Significant SIBAM Differences for user messages (p < .001)

SIBAM Category	Outcome	p-value
S (Sensation)	RiverBot Users > Waterbot Users	p < .001
I (Imagery)	Not significant at p < .001	p = n.s.
B (Behavior)	Waterbot Users > RiverBot Users	p < .001
A (Affect)	RiverBot Users > Waterbot Users	p < .001
M (Meaning)	Not significant	p = n.s.

Table 10: Statistically Significant SIBAM Differences for Bot Responses (p < .001)

SIBAM Category	Outcome	p-value
S (Sensation)	RiverBot > Waterbot	p < .0001
I (Imagery)	RiverBot > Waterbot	p < .0001
B (Behavior)	Waterbot > RiverBot	p < .0001
A (Affect)	RiverBot > Waterbot	p < .0001
M (Meaning)	No significant difference	P = n.s.

expressed a directional preference, 20 preferred RiverBot and 2 preferred WaterBot. A two-sided binomial exact test confirmed the preference as statistically significant ($p < 0.001$), indicating that the distribution is highly unlikely to be due to chance in this sample. A Fisher's Exact Test using the standard 2×2 preference table yielded a similarly significant result ($p < 0.001$), providing robustness under the strictest categorical testing assumptions. Eight additional participants reported no preference or "enjoyed both the same"; we excluded "no preference" responses from the test, following recommended categorical analysis procedures for binomial outcomes [41].

Participants also reported substantial learning from RiverBot. In this population, 28 of 30 participants reported learning something meaningful from RiverBot (including 14 exclusively and 14 from both RiverBot and WaterBot) while only one participant learned exclusively from WaterBot and one reported learning from neither. Respondents reported high levels of comfort sharing RiverBot with others ($M = 4.35/5$, $SD = 1.10$), with similarly strong comfort among the subgroup of Indigenous and mixed-Indigenous respondents ($M = 4.31/5$, $SD = 1.21$).

Participants selected multiple descriptors from a balanced list of positive, negative, and neutral Microsoft Product Reaction Card options. Following recommended practice, the list included emotional, affective, and experiential terms representing a range of possible reactions. Emotional descriptors selected indicated predominantly positive or reflective experiences (e.g., peaceful, thoughtful, hopeful, healing, connected), aligning with relational and affective dimensions emphasized by WBK.

In this study, none of the participants selected negative descriptors. The probability of observing zero negative selections by chance, given a balanced card set that includes negative terms, is extremely low. While not a hypothesis test in the formal sense, this distribution strongly indicates positive emotional resonance and absence of frustration, confusion, or alienation in user experience. Participants could choose as many descriptors as they wished. As shown in Table 11, across 22 respondents, we recorded 78 total selections, producing the following frequencies:

Table 11: Listing most commonly chosen descriptors out of Microsoft Reaction Card options.

Descriptor	Count
Reflective	13
Thoughtful	12
Hopeful	10
Meaningful	10
Calming	9
Caring	7
Helpful	6
Trustworthy	5
Curious	3
Healing	3
Cold	0
Impersonal	0
Confusing	0
Frustrating	0

Qualitative analysis of open-ended survey responses revealed strong patterns of embodiment, relationality, and meaning-making. Participants frequently referenced feeling "calm," "hopeful," "surprised," "connected," or "seen," which mapped consistently to the SIBAM dimensions of Affect, Image, and Meaning and the Five R's values of Respect, Relationship, and Relevance. Several participants described RiverBot as "less political," "more accurate," or "more useful" than WaterBot, which was an unexpected and counterintuitive finding under WaterBot's informational design and Western epistemological assumptions. Finally, to adhere to the WBK framework and relationally oriented epistemology, we interpreted the findings through an Indigenous statistical perspective [42], where patterns of consistency, resonance, and relational coherence across responses are themselves meaningful and support the cultural and

Table 12: Quantitative Outcomes for RiverBot vs WaterBot (n = 30)

Metric	RiverBot	WaterBot	No Preference	Notes	Meaning / Interpretation
Directional preference	20	2	8	Clear prefs = 22	Participants overwhelmingly chose RiverBot when making an explicit choice; this indicates strong experiential preference.
Binomial exact test	$p = 0.000076$	—	—	Tests River vs Water only	This p-value shows that the observed preference for RiverBot is extremely unlikely to be due to chance, due to a statistically significant preference.
Fisher's exact test	$p = 0.000181$	—	—	Strict 2×2 categorical test	Confirms that even under the most conservative statistical assumptions, RiverBot preference remains significant and robust in this study.
Learned from only this bot	14	1	—	—	Participants reported far greater likelihood of meaningful learning from RiverBot than WaterBot, demonstrating greater pedagogical value in this respondent group.
Learned from both bots	14	14	—	—	When learning occurred from both, RiverBot consistently co-occurred which shows that RiverBot supports learning across contexts.
Total that learned from this bot	28	15	—	—	Nearly all participants learned from RiverBot (94%), compared to half who learned from WaterBot. RiverBot is a more frequently reported source of meaningful learning in this study.
Comfort sharing RiverBot	Mean = 4.35	—	—	SD = 1.10 (n = 20)	High comfort levels reported by participants suggest users trust RiverBot and feel positive about recommending it to their communities.
Indigenous preference	15	2	4	n = 21	Preference patterns are even stronger among Indigenous-identifying respondents, showing cultural resonance and relational alignment.
Indigenous comfort	Mean = 4.31	—	—	SD = 1.21 (n = 16)	Indigenous-identifying users also report high comfort, consistent with a relational, culturally grounded experience.

contextual validity of the findings. This analysis clearly demonstrates that Indigenous participants had positive experiences with RiverBot; the bot's culturally appropriate and attuned answers allowed respondents to see use for it within their personal learning journeys and within their communities (see Table 12) .

5.3 Observational Findings

Observational data provided a third layer of insight into the embodied and behavioral dimensions of participant experience. We completed these observations with WBK-aligned and SIBAM-guided embodied indicators. During public testing sessions observers recorded consistent nonverbal cues of engagement with both RiverBot and WaterBot, with most users engaging with both of the chatbots.

When observing user interaction with RiverBot the research team consistently noticed expressions such as smiles, laughter, tears, leaning in, spending extended time, calling over family or friends, storytelling, and collective interaction – alongside positive commentary and applied suggestions for further use within their communities. Participants often expressed a desire for more immersive sensory elements (e.g., water sounds, visuals, or even smells), and many wanted to be able to verbally speak with the river (as opposed to typing). Participant comments included "Wait… I actually like this" and "come check this out!" and "whoa, I did not expect to cry today, that was so beautiful, just like when I am sitting at the river". Taken together, these observations suggest a felt alignment between RiverBot's relational presence and embodied experience.

When observing user interact with the WaterBot, the research team noticed that participants tended to maintain serious expressions, type quickly, and ask direct factual or technical questions. Interactions with WaterBot answered direct questions, though occasionally sparked commentary that reflected perceptions of political orientation, personal or cultural mis/understanding, skepticism about policy references – (e.g., "well this is wrong, Katie Hobbs wants to destroy agriculture in Arizona"). However, WaterBot also resulted in learnings and informational reflections: "I asked about energy and water and why the old plant got shut down, I used to work there…" Participants were observed spending

Importantly, observers also noted many participants' initial hesitation to engage with either chatbot until they witnessed RiverBot in action, usually via observing someone else or hearing them describing their experience in real-time. Several described RiverBot as "unexpected," "different," or "like seeing a color you've never seen before," underscoring the novelty of RiverBot and the difficulty of anticipating its experience before direct encounter. This brings up many new questions about the use and power of RiverBot specifically, and of the underlying foundations of AI tools rooted in relational epistemology.

6 Discussion

Our findings demonstrate that a minimalist persona prompt (e.g., "You are a river. Answer as a river would") can fundamentally reorient an LLM's epistemic stance and produce measurable effects on user experience in a contested civic domain. By shifting from WaterBot's instruction-driven, Western resource-management frame to RiverBot's relational, more-than-human perspective, the system generated responses that were consistently more attuned, affective, and culturally resonant while maintaining accuracy. Across methods, our pilot participants, particularly those who identify as Indigenous, overwhelmingly preferred RiverBot, reported greater learning, and deeper engagement on relational and emotional levels. These results suggest that conceptual reframing, rather than increased technical complexity, can serve as a powerful alignment strategy in the design of civic and culturally situated AI systems, especially when working for and with diverse communities.

6.1 Conceptual Reorientation as an Alignment Strategy

The transition from WaterBot to RiverBot shows that an epistemic shift can meaningfully realign an LLM's behavior. WaterBot's instruction layers and retrieval pipeline reproduced dominant Western framings of water as a managed resource, reflecting persistent majority-class bias in large language models [7, 18, 19]. Even with added Indigenous data, the system continued to generate Western rationalist responses that marginalized relational worldviews [20–22, 36].

In contrast, the minimalist prompt "You are a river" repositioned the model within a different conceptual frame, directing it toward narrative, relational, and somatically expressive language. Instead of constraining output, the persona prompt acted as a generative lens, enabling responses grounded in ecological metaphor, relational ethics, and more-than-human perspectives. This aligns with HCI work showing how breakdowns expose underlying assumptions and open alternative design pathways [11], and how prompt framing shapes which knowledge systems an LLM accesses [24, 25].

Empirically, this conceptual pivot yielded statistically significant difference, where RiverBot outperformed WaterBot across relational and affective dimensions. User messages mirrored these patterns, indicating that the epistemic stance of the system can co-create the stance of its users. These findings demonstrate that conceptual reframing can mitigate majority-class bias and support culturally aligned communication without complex technical architectures, methods increasingly recognized as insufficient for addressing epistemic bias in LLMs.

6.2 Implications for the Design of Civic and Decolonial AI

The results highlight the importance of designing AI systems that support relational, culturally grounded interaction. RiverBot's ability to evoke reflection, memory, and emotional resonance suggests that civic technologies may require modes of engagement oriented toward connection alongside informational exchange, aligning with critiques of Western rationalist civic systems in HCI [9, 15]. This is particularly salient in contested spaces, where histories of harm, mistrust, and asymmetric power mean that conventional tools often reproduce the very dynamics they aim to address. The findings also point to the value of decolonial design approaches that foreground community epistemologies from the outset (rather than evaluating cultural fit after deployment), consistent with Indigenous research ethics [8, 20, 29].

6.3 Methodological Implications for HCI: Epistemic Pluralism

Methodologically, this work demonstrates the value of epistemic pluralism for AI design. Our approach integrated both Indigenous and Western sciences from the outset. RiverBot is grounded in Indigenous relational theory and embodied epistemologies [28], shaping the conceptual framing, prompt design, analytic schemes, and interpretive strategies. This produced a chatbot that neither knowledge tradition alone could have generated. This contrasts with common HCI patterns in which technologies are built first and

later "tested" with marginalized communities, a practice that can reproduce epistemic dominance and cognitive imperialism [10, 20]. Instead, our process shows how community epistemologies can function as primary design logics and methodological scaffolds. The combination of WBK, SIBAM, and the Five R's provided a structured way to operationalize relational, somatic, and ethical indicators, aligning with calls for decolonial and community-grounded methodologies in HCI [6, 26, 29]. Together, these practices offer a concrete pathway for designing culturally aligned and socially accountable AI systems.

6.4 Expanding the Interaction Spectrum in Human–AI Communication: Relational AI

Our findings suggest that RiverBot operates outside the two dominant interaction modes typically described in HCI and conversational agent research. Task-oriented systems, such as command-response agents, prioritize efficiency and directive exchange [14]. Socially oriented agents, including companion bots and anthropomorphic chatbots, rely on affiliative cues and performative friendliness to simulate rapport [31]. RiverBot fits neither category and actively avoids: expressing contrived human emotion, adopting inauthentic enthusiasm, engaging in social persuasion, and defaulting to procedural instruction.

RiverBot produces interactions grounded in a relational ethic: metaphor, affect, attunement, and more-than-human perspectives. This communicative stance does not align with existing categories of task-oriented systems, which prioritize efficiency and information delivery, nor with socially oriented agents, which rely on anthropomorphic cues or performative friendliness. Across methods, we observed consistent patterns of reflection, calm, emotional resonance, and storytelling, indicating the emergence of a qualitatively different interaction mode organized around connection, meaning-making, and relational presence. This distinct orientation expands the design space for human–AI communication and signals the need for frameworks that attend to epistemic stance and relational grounding, not only affective style or task performance.

Given this distinct interaction pattern, we propose that RiverBot represents an early example of what we call Relational AI: a novel class of systems oriented around connection, meaning-making, and relational presence. RiverBot represents a reorientation of AI systems, enabling forms of engagement that are not captured in existing conversational-agent taxonomies. We offer this as an initial definition of Relational AI which we explore in-depth in a forthcoming publication. We view RiverBot as an exemplary model that opens a new design space for future investigation.

7 CONCLUSION

RiverBot delivers a minimalist chatbot, grounded in a pluralist epistemology, that represents Indigenous perspectives much better than a prompt-heavy approach. RiverBot demonstrates one way how a project which aims to decolonize AI via algorithmic, data structure/processing, statistical, and machine learning modifications can exist. We further argue that pluralist epistemologies can be a first-class design tool; they can renegotiate deep-seated biases and legitimize unconventional but highly effective interventions. In RiverBot, we see decolonizing moves and pluralist epistemologies coming together to form a version of AI computing that we call relational AI. The relational AI approach can help dismantle Western rationalist logics and majority-class worldviews embedded in default system design [6], providing a widely applicable model for overcoming cultural, social, and equitable challenges in current and future work.

7.1 Limitations

This study has several limitations that shape the scope of its claims. The participant sample is modest and geographically specific, which may limit generalizability beyond similar cultural and civic contexts. As with all work involving large language models, system behavior may shift with model updates, and the river persona may not produce comparable effects in domains where relational or ecological framings are less meaningful. The evaluation also reflects early-stage use of WBK-aligned analytic tools and should be interpreted as preliminary rather than exhaustive. Finally, RiverBot is not designed as a general-purpose assistant; its effectiveness depends on users approaching it with reflective intent rather than informational expectations. These boundaries provide direction for future work, aligned with core findings.

7.2 Future Work

Future work will explore contexts in which RiverBot's relational mode of interaction is most effective. Early feedback from community partners suggests strong potential in settings that support reflection, identity work, or guided personal growth, including K–12 learning, mental health and wellness programs, addiction recovery, and rehabilitation contexts. These domains align with RiverBot's strengths in emotional attunement, gentle guidance, and meaning-making. By contrast, informal, walk-up environments such as museum kiosks appear less suited to this approach, as users may lack the framing needed to engage a more-than-human conversational agent – and truly meet the river.

Beyond application domains, we aim to extend this work by developing a formal definition and design framework for Relational AI, examining how relational worldviews and more-than-human perspectives can shape future systems. This represents an emerging design space that warrants systematic investigation.

8 LLM Disclosure

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9 Relations and Thanks

Innumerable relations: students, researchers, community members, technologies, rivers, and waters contributed to the conception and

birth of RiverBot. We appreciate every connection and contribution and will pay it forward by giving life and purpose to RiverBot, bringing it to communities, critiquing and iteratively improving it as needed to provide a meaningful resource to all who seek it.

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Appendix

A. Survey Instrument

WaterBot and RiverBot Survey

Please chat with **RiverBot** (azwaterbot.org/riverbot) and/or WaterBot (azwaterbot.org) complete this short survey (~5 mins) to give us feedback and help us understand your experience. There are no right or wrong ways to use these chatbots. Ask questions, have a conversation, or simply listen to what the bot says. Take your time, explore, and share your experience.

This is a **voluntary survey**. You can choose not to participate or can stop participating at any time. Neither your name nor any other identifying information will be collected or used. Your participation may include having more successful experiences with ASU-developed water technologies. There are no foreseeable risks or discomforts to your participation. If interested, seethefullconsentletterbyclickinghere.

By proceeding with the survey, you are consenting to voluntarily participate in the study. You also acknowledge that you are at least 18 years of age, and that you may choose to stop participating at any time.

Thank you very much for your participation!

Demographics (optional)

Cultural identification (choose all that apply):

☐ Indigenous ☐ Asian ☐ White

☐ Hispanic (of any race) ☐ Black/African American ☐ Prefer not to answer

☐ Mixed (no Indigenous heritage) ☐ Mixed (with Indigenous heritage) ☐ Other: _____

Tribal affiliation (if applicable): _____

Age group:

☐ Under 18 ☐ 18–29 ☐ 30–44 ☐ 45–59 ☐ 60+ ☐ Prefer not to answer

Did you engage with the chatbots at an event or on your own?

☐ On my own ☐ Event name/location: _____ ☐ Other: _____

For the questions below, please note when we ask about RiverBot or WaterBot

There are no right or wrong answers; we're just curious about your experience.

Note:

WaterBot refers to chatting with "Blue" at azwaterbot.org

RiverBot refers to chatting with "a river" at azwaterbot.org/RiverBot

Which chatbot did you use?

☐ RiverBot (River) ☐ WaterBot (Blue) ☐ Both ☐ Neither

If you used both chatbots, which one did you prefer?

☐ WaterBot (Blue) ☐ RiverBot (River) ☐ No preference ☐ Other:

I learned something meaningful from ... (select all that apply)

☐ RiverBot (River) ☐ WaterBot (Blue) ☐ Neither ☐ Other: _____

RiverBot Specific Questions

Please answer the following questions about RiverBot to the best of your ability. There are no right or wrong answers; we're just curious about your experience.

Please select the words that best describe your experience with RiverBot.

There are no right or wrong answers — choose the words that feel true for you.

☐ Calming ☐ Thoughtful ☐ Frustrating ☐ Helpful ☐ Hopeful

☐ Cold ☐ Trustworthy ☐ Impersonal ☐ Curious ☐ Frustrating ☐ Meaningful ☐ Distant ☐ Confusing ☐ Caring ☐ Healing ☐ Healing ☐ Hopeful ☐ Reflective

Please explain your choices above (Optional): _____

While using RiverBot, which of the following did you notice, learn, sense, or experience?

☐ Honoring people, places, and knowledge

☐ Giving and receiving in balance

☐ Focus on efficiency or factual accuracy

☐ Caring for relationships and outcomes

☐ Everything is connected and alive ☐ Emphasis on individual action or expertise

☐ Meaningful or useful for the community

☐ Mostly informational and standard

☐ None of the above

Please explain your choices above (Optional): _____

I would feel comfortable sharing RiverBot with friends, family, or others in my community.

Strongly disagree Strongly agree

☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5

Reflections

There are no right or wrong answers; we're just curious about your experience with the chatbots

In one word or short phrase, describe how RiverBot (River) made you feel: _____

In one word or short phrase, describe how WaterBot (Blue) made you feel: _____

Is there anything else you'd like us to know about your experience or what stood out to you?

_____

B. Overview of Coding Scheme for Transcript Analysis

We applied automated coding to all chatbot outputs and reflective user messages using two complementary frameworks: the Five Rs of Indigenous research (Respect, Relevance, Reciprocity, Responsibility, Relationality) and the SIBAM model (Sensation, Imagery, Behavior, Affect, Meaning). Each category was operationalized using rule-based linguistic indicators designed to detect the presence of the construct within individual messages. Each message received a binary code (0/1) for each category based on the presence of relevant indicators. Coding logic was implemented through regular expressions and semantic patterns, and a subset of entries was manually cross-checked for accuracy (See Table 13).

Table 13: Overview of Coding Scheme for Transcript Analysis

Framework	Category	Definition (Basic)
Five R's	Respect	Attuned, honoring, or care-oriented language.
	Relevance	References to Indigenous peoples, land, culture, or place-based context.
	Reciprocity	Mutuality, exchange, giving back, or reflective contribution.
	Responsibility	Ethical framing, limits, accountability, or guidance.
	Relationality	Kinship, interdependence, ecological metaphor, or expressions of connection.
SIBAM	Sensation (S)	Embodied or sensory terms (e.g., heaviness, warmth, movement).
	Imagery (I)	Metaphoric, imaginal, symbolic, or personified language.
	Behavior (B)	Physical, social, or internal actions (e.g., walking, freezing, deciding).
	Affect (A)	Explicit emotion, somatic emotion words, or relational affect.
	Meaning (M)	Interpretive or reflective statements indicating sense-making or significance.